

Practice Question: The Allais Paradox

Axioms, Geometry, and Experimental Evidence

Professor Daeyoung Jeong

Yonsei University

Experimental Setup

A Laboratory Experiment

A decision-maker faces two pairwise choice problems over lotteries with real monetary payoffs. The outcome space is $X = \{\$50, \$15, \$0\}$.

Decision Problem 1: Choose between A and B .

- A : \$15 with certainty.
- B : \$50 with prob. 0.20, \$15 with prob. 0.75, \$0 with prob. 0.05.

Decision Problem 2: Choose between A^* and B^* .

- A^* : \$15 with prob. 0.25, \$0 with prob. 0.75.
- B^* : \$50 with prob. 0.20, \$0 with prob. 0.80.

Empirical Result: Suppose 68% of subjects choose A in Problem 1 *and* B^* in Problem 2. This (A, B^*) pattern is the modal choice and is inconsistent with Expected Utility Theory.

Part A: The Axiomatic Violation

- **Question A.1:** Assume a decision-maker follows Expected Utility Theory (EUT) with Bernoulli utility $u(\cdot)$. Show mathematically that a decision-maker who strictly prefers A to B must also strictly prefer A^* to B^* .
- **Question A.2:** Identify the axiom of EUT that is tested and violated by the (A, B^*) pattern. State this axiom formally for any three lotteries L, L', L'' and mixing probability $\alpha \in (0, 1)$, and show how the four lotteries in this experiment fit its structure.

Part B: Graphical Illustration (Marschak–Machina Triangle)

- **Context:** Represent this experiment in the Marschak–Machina probability triangle for the three fixed outcomes $x_H = \$50$, $x_M = \$15$, $x_L = \$0$.
- **Question B.1:** State which probabilities are measured on the horizontal axis and on the vertical axis of the triangle. Compute the coordinates (p_L, p_H) of all four lotteries (A, B, A^*, B^*) , and compute the slopes of the line segments $\overline{AB}$ and $\overline{A^*B^*}$.
- **Question B.2:** Using these slopes and the shape of EUT indifference curves in the triangle, explain why the modal (A, B^*) pattern cannot be generated by any expected-utility maximizer. Illustrate your answer with a diagram.

Part C: Behavioral Implications

- **Context:** The systematic violation of EUT in this design suggests that individuals do not evaluate lotteries linearly in probabilities.
- **Question C.1:** What psychological effect explains this pattern, and how does it change our view of what drives risk attitudes?
- **Question C.2:** Non-expected utility theories capture this with a probability weighting function $w(p)$. State the mathematical properties and boundary conditions w must satisfy on $[0, 1]$. Then, using $V(L) = \sum_i w(p_i) v(x_i)$ with $v(0) = 0$, show which condition on w is required to generate the (A, B^*) pattern.